PROGRAMMING IN OPHTHALMOLOGY · SESSIONS 1–2

ML for clinicians – cheat sheet

ONE PAGE · KEEP IT NEXT TO YOUR KEYBOARD

1 · Before you write any code

- **Question:** which clinical decision would this support?
- **Unit of analysis:** what is ONE row – eye, visit, patient, image?
- **Label:** who defined it, when, how – independent of the features?
- **Features:** available at the moment of use? Nothing from the future?
- **Split:** by patient (or site, or time) – never by row.
- **Metric & baseline:** sensitivity/specificity vs. "always predict the majority".
- **Failure analysis:** subgroups, correlated features, circular labels.
- **Before clinical use:** external validation, prospective test, regulation.

2 · Data handling

- Compare **rows vs. unique patients vs. unique eyes** first.
- Merge **by key**, never by row position; let pandas check: `validate="one_to_one"`.
- Clean with **explicit, logged decisions** (what was removed, why, how many) – this table goes into Methods.
- Check codes of the label, missing values, impossible values, duplicates / doublets.
- Know your **sign conventions**: Octopus MD positive = loss, Humphrey MD negative = loss.

```
data = a.merge(b, on=KEY, how="left",
               validate="one_to_one")
```

3 · Leakage – five faces to look for

- **Same patient on both sides** of the split (both eyes, repeat visits, many images).
- **Preprocessing before the split** (scaling, imputation, feature selection on ALL data).
- **Features that contain the answer** / label defined from the features (circular).
- **Duplicates & doublets** (same person under two IDs).
- **Time and place**: train and test from the same clinic/device/period.

Ask of any paper: what was independent between training and test – and was it enough?

4 · Split, scale, validate

```
from sklearn.model_selection import (
    GroupShuffleSplit, GroupKFold)
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression

split = GroupShuffleSplit(test_size=.25, random_state=42)
tr, te = next(split.split(X, y, groups))  # patient IDs
pipe = make_pipeline(StandardScaler(),    # fit on TRAIN
                     LogisticRegression())
for tr, te in GroupKFold(10).split(X, y, groups):
    pipe.fit(X.iloc[tr], y.iloc[tr]); ...
```

- Trees do not need scaling; logistic regression / NN do.
- One split = one draw → use **patient-wise cross-validation**.
- Tuning on the folds you report = mild leakage (use nested CV / a final untouched test set).

5 · Metrics from the confusion matrix

	predicted +	predicted –
truly +	TP	FN (missed)
truly –	FP (false alarm)	TN

Metric	Formula	Question
Sensitivity	TP/(TP+FN)	cases caught?
Specificity	TN/(TN+FP)	healthy cleared?
PPV	TP/(TP+FP)	positive = real? *(depends on prevalence!)*
NPV	TN/(TN+FN)	negative = healthy?
Accuracy	(TP+TN)/all	hides error types
AUC	area under ROC	0.5 coin flip · 1.0 perfect

6 · Thresholds, uncertainty, reporting

- **Threshold** = clinical decision. Low → catches more cases, more false alarms (screening). High → fewer false alarms, more misses (confirmatory).
- **Bootstrap CI:** resample **patients** (not rows) with replacement ≈1,000×, take 2.5th–97.5th percentile.
- Small test sets → wide intervals. Overlapping intervals → no proof that one model is better.
- **Report:** metric + 95 % CI + how the split was made + baseline + study population.

"On N unseen patients (patient-wise split), sensitivity was X % (95 % CI a–b), specificity Y % (c–d); baseline accuracy Z %."

7 · Red flags – stop and ask questions

- **Too good to be true** (accuracy/AUC ≈ 1.0) → leakage or a label defined from the features.
- **One feature separates the groups** → how was the label defined?
- **Coefficients jump** when a correlated feature is added/removed → interpret groups of features.
- **"Important to the model"** ≠ important to the eye; association ≠ causation.
- **Model built in a clinic** used for screening → prevalence (PPV) and population differ.
- **Only one split / no baseline / no CI** → the number is an anecdote.

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github.com/simonkng/ophthalmology-ml-workshop